Executive Summary

E-commerce Customer Intelligence & Cancellation Risk Analytics

---

1)Business Context

The rapid growth of e-commerce platforms has generated massive amounts of transactional and customer behavior data. Analyzing this data effectively can help businesses improve customer retention, optimize operational performance, and reduce revenue losses caused by order cancellations.

This project focuses on analyzing an online retail dataset to uncover customer purchasing patterns, identify valuable customer segments, and predict cancellation risks using business intelligence and machine learning techniques. The goal is to transform raw transactional data into actionable insights that support data-driven decision-making.
The project combines data cleaning and transformation in Excel, interactive dashboard development in Power BI, and predictive analytics using Python-based machine learning models.

2)Dataset & Methodology

The project utilized the Online Retail II dataset, which contains transactional records from UK-based online retail company. The dataset includes customer purchases, product details, invoice information, quantities, prices, transaction dates, and cancellation records.

Because the dataset was originally raw and inconsistent, a comprehensive data preparation process was conducted primarily using Microsoft Excel. This phase included handling missing customer records, separating cancelled transactions, correcting data formats, removing inconsistencies, and preparing the data for analysis.

To support deeper business analysis, additional calculated fields were created, including total salses value, purchase hour, day of the week, and weekend indicators. These feature helped improve trend analysis, customer behavior analysis, and predictive modeling .

After the preparation phase, the cleaned dataset was imported into Power BI to build interactive dashboards for sales performance analysis, customer segmentation, trend monitoring, and cancellation analysis .
For advanced analytics, Python-based machine learning techniques were applied, RFM analysis and K-Means clustering were used to segment customers into behavioral groups, while a Random Forest classification model was developed to predict cancellation risk and identify the most influential predictive factors.

3) Key Insights

The analysis generated several important insights regarding customer behavior, sales performance, and cancellation patterns within the e-commerce environment.

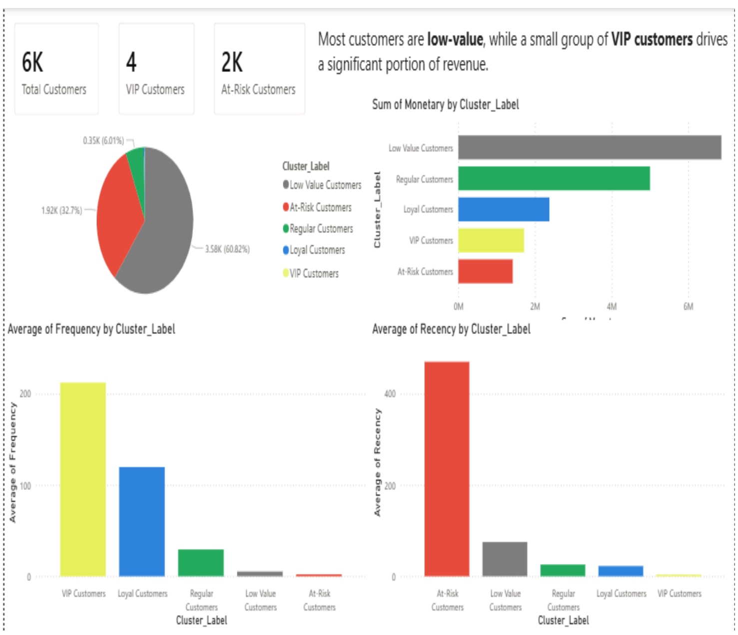
The business generated over 17 million in total revenue across approximately 37 thousand orders and 6 thousand customers. Sales activity showed strong temporal behavior, with purchasing activity peaking around midday, particularly at 12 PM, while weekday sales significantly outperformed weekend sales. Thursday was identified as the highest revenue-generating day.

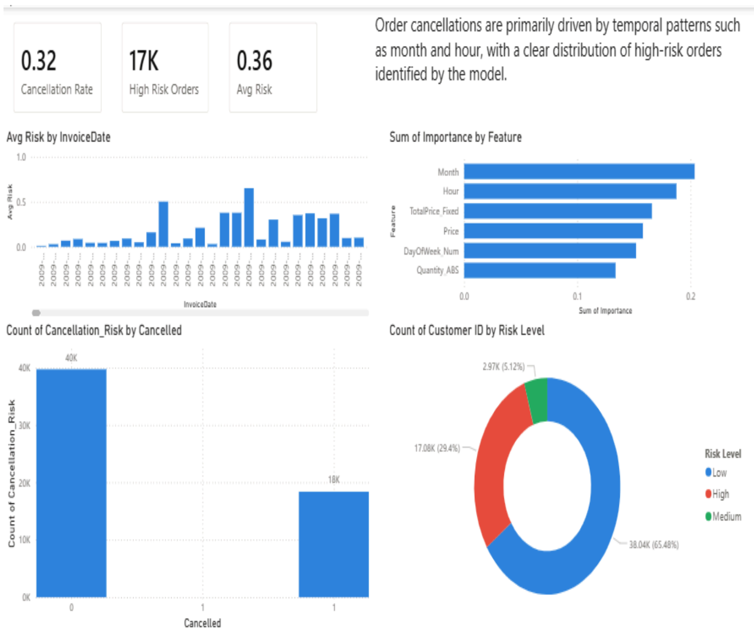
Customer segmentation analysis revealed that the majority of customers belonged to low-value segments, while a relatively small group of VIP and Loyal customers contributed a significant share of total revenue. Using RFM analysis and K-Means clustering, customers were categorized into behavioral groups including VIP Customers, Loyal Customers, Regular Customers, At-Risk Customers, and Low-Value Customers. The segmentation model achieved a silhouette score of approximately 0.59, indicating a reasonably strong separation between customer groups.

Cancellation analysis demonstrated that cancellation activity increased toward the end of the year, reaching its highest levels during December. Additionally, the United Kingdom accounted for the majority of both revenue generation and cancelled orders, while specific products contributed disproportionately to cancelled revenue.

For predictive analytics, a Random Forest classification model was developed to estimate cancellation risk. Before model training, class balancing techniques were applied to address dataset imbalance between cancelled and non-cancelled transactions. The model achieved an accuracy of approximately 71%, demonstrating a balanced ability to identify both cancelled and non-cancelled transactions.

Feature importance analysis showed that month, purchase hour, pricing behavior, and order quantities were among the strongest predictors of cancellations. These findings suggest that temporal and transactional patterns play a significant role in cancellation behavior within e-commerce operations.





4) Recommendations

Based on the analysis results, several business recommendations can be proposed to improve operational performance and customer retention.

The company should prioritize retaining VIP and Loyal customers through personalized offers, loyalty programs, and targeted engagement campaigns, as these customer groups contribute significantly to overall revenue generation.

Since cancellation activity increases during specific periods, particularly toward the end of the year, the business should closely monitor operational performance during high-risk periods and investigate the causes of increased cancellations. Predictive cancellation risk analysis can also be integrated into operational workflows to identify risky transactions early and support proactive decision-making.

The organization may further improve marketing effectiveness by using customer segmentation insights to create personalized campaigns, product recommendations, and retention strategies tailored to different customer groups.

Additionally, temporal purchasing patterns suggest that businesses should optimize staffing, inventory management, and promotional activities during peak purchasing hours and high-performing weekdays to maximize operational efficiency and revenue generation.

5) Limitations

Despite the valuable insights generated, the project has several limitations.

The dataset represents historical transactions from a single online retail environment, which may limit the generalizability of the findings to other industries or business models. Some customer records also contained missing or incomplete information, requiring filtering and cleaning during the preparation phase.

The predictive model was developed using historical transactional behavior and therefore cannot guarantee perfect future predictions. External factors such as customer preferences, economic conditions, marketing campaigns, and operational changes were not included in the analysis and may influence future outcomes.

In addition, the customer segmentation process depends on behavioral assumptions derived from RFM analysis and clustering techniques, meaning customer groups may evolve over time as purchasing behavior changes.